

Explanation of Transcript of Records

– Course Director's Statement

Mr. Anton Kühr is currently studying the **innovative bachelor's** degree programme "Mechanical Engineering – Product Engineering and Context (MPEC)" at Cologne University of Applied Sciences. The MPEC curriculum follows an **integrated, competence-oriented learning concept that combines scientific, technical, interdisciplinary and reflective components** into coherent modules rather than separate, purely lecture-based courses. As a result, many of the **module titles in this transcript of records may not be self-explanatory** or directly comparable to more traditional curricula. An **overview of the structure and content** of these modules is **provided at the end of this document**. Should any questions remain regarding the content, scope, or academic level of these courses, please feel free to contact me directly.

Furthermore, in the first three semesters, each semester is organised around a single, **integrated project module worth 30 ECTS**. These project modules are **assessed on a pass or fail** basis without differentiated grades. Consequently, a **qualified performance evaluation is provided after the transcript of records**.



27.11.2025

Prof. Dr.-Ing. Stefan Benke (Course Director MPEC)

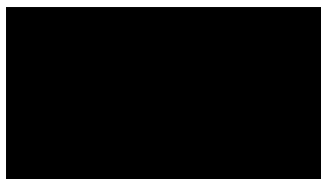
Prof. Dr.-Ing. Stefan Benke
Engineering Mechanics and
Mechanics of Material

Head of Degree Programme
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- Product Engineering
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Transcript of Records

Anton Karl Ludwig Kühr
Degree program: Mechanical Engineering - Product Engineering and Context
Intended degree: Bachelor's degree

Date of birth:
Place of birth: Köln

Exam No.	Course	Semester	Grade	Credits	Annot.
974	ECTS creditis	SS 26	1.1	210	
955	Prerequisites for thesis completed	SS 26		0	
952	Bachelor's thesis	SS 26	1.0	12	
950	Bachelor's thesis Investigation of the Influence of Mechanical Microphone Implementation on ASR Performance	SS 26	1.0	12	Actual submission date: March 27, 2026
962	Final oral examination	SS 26	1.0	3	
960	Final oral examination	SS 26	1.0	3	
1100	Engineering Office I	WS 22/23		30	
1010	Engineering Office I	WS 22/23		30	
2100	Engineering Office II	SS 23		30	
2010	Engineering Office II	SS 23		30	
3100	Engineering Office III	WS 23/24		30	
3010	Engineering Office III	WS 23/24		30	
4000	Examinations semesters 4+5	SS 25		56	
4100	Product Engineering I	WS 24/25	1.0	10	
4110	Product Engineering I	SS 24	1.0	10	

Mr. Anton Karl Ludwig Kühr born on [REDACTED] in Köln Student ID : [REDACTED]
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Exam No.	Course	Semester	Grade	Credits	Annot.
4119	Product Engineering II	WS 24/25	1.0	10	
4120	Product Engineering II	WS 24/25	1.0	10	
4200	Reflection 4	SS 25	1.3	3	
4205	Reflection 4	SS 25	1.3	3	
4209	General	SS 25	1.3	5	
4210	General	SS 25	1.3	5	
4211	Control Engineering	SS 24	1.0	5	
2320	Control Engineering	SS 24	1.0	5	
4212	Engineering Mathematics	SS 24	1.6	5	
4320	Engineering Mathematics	SS 24	1.6	5	
4221	Interdisciplinary Product Design Project	WS 24/25	1.7	5	
4220	Interdisciplinary Product Design Project	WS 24/25	1.7	5	
4222	Engineering Thermodynamics	WS 24/25	2.0	5	
1515	Engineering Thermodynamics	WS 24/25	2.0	5	
4223	Energy System Simulation	WS 24/25	1.7	5	
4420	Energy System Simulation	WS 24/25	1.7	5	
4500	Reflection 5	SS 25	1.0	3	
4505	Reflection 5	SS 25	1.0	3	
5000	Required Electives	SS 25		10	
5200	Social Robotics 1: Human-Machine-Interaction (MO)	SS 25	1.7	5	
5210	Social Robotics 1: Human-Machine-Interaction (MP)	SS 24	1.7	5	
5250	Social Robotics 2: Service and Industry Robotics (MO)	SS 25	1.7	5	
5220	Social Robotics 2: Service and Industry Robotics	SS 25	1.7	5	

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Exam No.	Course	Semester	Grade	Credits	Annot.
6000	Examinations semester 6	WS 25/26		30	
6100	Engineering Lab	WS 25/26		25	
6105	Engineering Lab	WS 25/26		25	
6200	Reflection 6	WS 25/26	1.0	5	
6205	Reflection 6	WS 25/26	1.0	5	
7000	Examinations semester 7	SS 26		9	
7050	Proposal	WS 25/26	1.0	5	
7051	Proposal	WS 25/26	1.0	5	
7100	Symposium	SS 26	1.0	2	
7110	Symposium	SS 26	1.0	2	
7200	Reflection 7	WS 25/26	1.0	2	
7205	Reflection 7	WS 25/26	1.0	2	

This Transcript of Records is valid without signature.

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Explanations

Grading scheme: 1-1.5 = excellent; 1.6-2.5 = good; 2.6-3.5 = satisfactory; 3.6-4.0 = sufficient; >4.0=fail

Annotations: ANA=medical certificate not recognized ANB=mandatory registration AAK/AEP=credit transfer CNB=Corona free examination attempt not passed EAA=Thesis submitted electronically FDA=submission deadline of Diplom Thesis exceeded MEP=additional oral examination NÄN=change of grade NAN=late registration NPA=grade given by the Chairperson of the Examination Board NZU=not allowed PNE=did not appear for exam PFV/PZW=potential free examination attempt SOA=special request TR=withdrawal from thesis UBE=graded examination passed but no grade given UEB=over subscribed UMB=transfer UR=unexcused withdrawal TUH=attempted cheating VTW=change of specialization ZWW=mandatory registration

Other: SS=summer semester WS=winter semester

Supplementary module description

Mechanical Engineering - Product Engineering and Context (MPEC)

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Engineering Office I (30 ECTS credits)

The Engineering Office 1 module is an interdisciplinary project module in which a **smart mechatronic system, using concepts from systems engineering, is analysed**. In the course of the project, students acquire **fundamental knowledge of mathematics, engineering mechanics, physics, materials science, computer science and reflection**. These contents correspond to those taught in traditional engineering courses at Cologne University of Applied Sciences and, for each subject area, represent a workload of 5 ECTS, matching the technical depth of comparable first-semester modules.

Evaluation method: passed or failed

Engineering Office II (30 ECTS credits)

The Engineering Office 2 module is an interdisciplinary project module in which students **design, build and commission a simple smart mechatronic system**. In the course of the project, students **deepen their knowledge of mathematics, technical mechanics, materials science, computer science, design engineering and reflection**. These contents correspond to those taught in traditional engineering courses at Cologne University of Applied Sciences and, for each subject area, represent a workload of 5 ECTS, matching the technical depth of comparable second-semester modules.

Evaluation method: passed or failed

Engineering Office III (30 ECTS credits)

The Engineering Office 3 module is an interdisciplinary project module in which students **design, build and commission an advanced smart mechatronic system**. In the course of the project, students **deepen their knowledge of technical mechanics, computer science, design engineering, product development, project organisation and reflection**. These contents correspond to those taught in traditional engineering courses at Cologne University of Applied Sciences and, for each subject area, represent a workload of 5 ECTS, matching the technical depth of comparable third-semester modules.

Evaluation method: passed or failed

Product Engineering I (10 ECTS credits)

The aim of the Product Engineering 1 module is to **develop and answer an *initial, simple scientific question***. Students can either develop a new product or improve an existing product in terms of sustainability. The **structured product development process** should be followed and the **sustainable improvement of the product** should be presented. **Alternatively, a technical question can be analysed using scientific research methods**. Here, the developed mechatronic device is a tool for investigating the question. The collection, evaluation and discussion of the data is the main part of the work.

Evaluation method: grade according to the German grading system

Product Engineering II (10 ECTS credits)

The aim of the Product Engineering 2 module is to **develop and answer an *advanced scientific question***. Students can either develop a new product or improve an existing product in terms of sustainability. The **structured product development process** should be followed and the **sustainable improvement of the product** should be presented. **Alternatively, a technical question can be analysed using scientific research methods**. Here, the developed mechatronic device is a tool for investigating the question. The collection, evaluation and discussion of the data is the main part of the work.

Evaluation method: grade according to the German grading system

General (5 ECTS credits)

The General course (Studium generale) allows students to choose a subject from the wide range of courses offered by Cologne University of Applied Sciences. It enables students to **learn about other disciplines beyond their own field of study and to think in an interdisciplinary way**. It serves the purpose of supporting personal and academic development. The programme often includes lectures, discussions, seminars and cultural events to broaden horizons and open up new perspectives.

Evaluation method: grade according to the German grading system

Interdisciplinary Design Project (5 ECTS credits)

In the Interdisciplinary Design Project module, students **examine how technical products perform in terms of user-friendliness, accessibility, sustainability and social acceptance**. To do this, they design or further develop products and user interfaces in practical projects and then evaluate them with regard to usability and sustainability.

Evaluation method: grade according to the German grading system

Reflection 4-7 (2-5 ECTS credits)

In the Reflection modules, students systematically examine their project work and teamwork. In small groups, they continuously reflect throughout the semester on both subject-specific and interdisciplinary aspects of their joint work. The aim of the modules is to **strengthen their ability to engage in self-directed reflection and to raise awareness of the importance of individual and group contributions to the success of technical projects**, both at a technical level and with regard to team processes. Students learn to use the developed criteria matrix to evaluate their own performance and that of their fellow students in a sound, well-reasoned manner and to give constructive feedback.

Evaluation method: grade according to the German grading system